



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-31204

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Tue, Aug 27, 2024 at 1:44 AM

Reply-To: arianna.dulizia@irpps.cnr.it

To: khare.pranav@gmail.com

27-Aug-2024

Dear Mr. Khare:

The manuscript entitled "Exploring Collaboration between Chain-of-Thought-Tuned Conceptual Design Agents and Designers: A Comparative Case Study" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Dr. Arianna DULizia

IEEE Access Associate Editor
arianna.dulizia@irpps.cnr.it

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Exploring Collaboration between Chain-of-Thought-Tuned Conceptual Design Agents and Designers: A Comparative Case Study

AUTHORS: Ge, Shijun; Sun, Yuanbo; Cui, Yin; Wei, Dapeng

ABSTRACT: This study aims to explore the cognitive impact of agents based on the large language model GPT-4 that are fine-tuned through a chain-of-thought (CoT) approach on conceptual design collaboration. By fine-tuning agents via traditional brainstorming (IntelliStorm) and structured TRIZ (EvoluTRIZ) methods as well as utilizing electrodermal activity (EDA) physiological data and expert scoring, this study evaluates these agents' influence on designers' emotional activation and innovative thinking abilities. The experimental results indicate that agents that are fine-tuned via different conceptual design methods significantly affect the task processes and outcomes of design participants. Participants exhibited notable differences in emotional responses during the task. The IntelliStorm agent exhibited superior performance with regard to enhancing thought fluency, whereas the EvoluTRIZ agent offered significant advantages in terms of originality and elaboration. This study provides empirical foundations for the construction of agents via traditional design methods based on CoT technology and offers insights for further agent optimization. The findings offer not only practical implications for the design field but also new perspectives on design education, which emphasize the importance of cultivating thinking abilities. Future research can explore the application of agents in various design domains and optimize the interaction patterns between agents and humans in collaborative settings

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